

Problem

Find the inverse Fourier transforms of the following functions:

$$(a) \quad F(\omega) = \frac{100}{j\omega(j\omega + 10)}$$

$$(b) \quad G(\omega) = \frac{10j\omega}{(-j\omega + 2)(j\omega + 3)}$$

$$(c) \quad H(\omega) = \frac{60}{-\omega^2 + j40\omega + 1300}$$

$$(d) \quad Y(\omega) = \frac{\delta(\omega)}{(j\omega + 1)(j\omega + 2)}$$

Step-by-step solution

Step 1 of 12

(a)

Consider the Fourier transform $F(\omega)$ as,

$$F(\omega) = \frac{100}{j\omega(j\omega + 10)}$$

Determine the inverse Fourier transform of $F(\omega)$.

$$\begin{aligned} F^{-1}[F(\omega)] &= F^{-1}\left[\frac{100}{j\omega(j\omega + 10)}\right] \\ \left[\frac{100}{j\omega(j\omega + 10)}\right] &= \frac{A}{j\omega} + \frac{B}{(j\omega + 10)} \\ &= \frac{A(j\omega + 10) + B(j\omega)}{j\omega(j\omega + 10)} \\ &= \frac{j\omega(A + B) + 10A}{j\omega(j\omega + 10)} \end{aligned}$$

Step 2 of 12

Compare constant terms on both sides.

$$10A = 100$$

$$A = 10$$

Compare $j\omega$ terms on both sides.

$$(A + B) = 0$$

$$A = -B$$

$$B = -10$$

$$\left[\frac{100}{j\omega(j\omega + 10)}\right] = \frac{10}{j\omega} - \frac{10}{(j\omega + 10)}$$

Step 3 of 12

Consider,

$$F^{-1}[F(\omega)] = F^{-1}\left[\frac{100}{j\omega(j\omega+10)}\right]$$

$$\begin{aligned} f(t) &= F^{-1}\left[\frac{10}{j\omega} - \frac{10}{(j\omega+10)}\right] \\ &= F^{-1}\left(\frac{10}{j\omega}\right) - F^{-1}\left(\frac{10}{(j\omega+10)}\right) \end{aligned}$$

Consider the Fourier transform of $\text{sig}(t)$.

$$F^{-1}\left(\frac{2}{j\omega}\right) = \text{sig}(t)$$

$$\begin{aligned} F^{-1}\left(\frac{10}{j\omega}\right) &= F^{-1}\left(\frac{5 \times 2}{j\omega}\right) \\ &= 5 \left[F^{-1}\left(\frac{2}{j\omega}\right) \right] \\ &= 5 \text{sig}(t) \end{aligned}$$

Step 4 of 12

Consider the Fourier transform of exponential decay function.

$$\begin{aligned} F^{-1}\left(\frac{1}{(j\omega+a)}\right) &= e^{-at}u(t) \\ F^{-1}\left(\frac{10}{(j\omega+10)}\right) &= 10F^{-1}\left(\frac{1}{(j\omega+10)}\right) \\ &= 10e^{-10t}u(t) \\ f(t) &= F^{-1}\left(\frac{10}{j\omega}\right) - F^{-1}\left(\frac{10}{(j\omega+10)}\right) \\ &= 5\text{sig}(t) - 10e^{-10t}u(t) \end{aligned}$$

Therefore, the Inverse Fourier transform of $F(\omega)$ is $\boxed{5\text{sig}(t) - 10e^{-10t}u(t)}$.

Step 5 of 12

(b)

Consider the Fourier transform $F(\omega)$ as,

$$F(\omega) = \frac{10j\omega}{(-j\omega+2)(j\omega+3)}$$

Determine the inverse Fourier transform of $F(\omega)$.

$$\begin{aligned} F^{-1}[F(\omega)] &= F^{-1}\left[\frac{10j\omega}{(-j\omega+2)(j\omega+3)}\right] \\ \left[\frac{10j\omega}{(2-j\omega)(j\omega+3)}\right] &= \frac{A}{(2-j\omega)} + \frac{B}{(j\omega+3)} \\ &= \frac{A(j\omega+3) + B(2-j\omega)}{(2-j\omega)(j\omega+3)} \\ &= \frac{j\omega(A-B) + (3A+2B)}{(2-j\omega)(j\omega+3)} \end{aligned}$$

Step 6 of 12

Compare constant terms on both sides.

$$3A + 2B = 0$$

$$3A = -2B$$

$$A = \frac{-2B}{3}$$

Compare $j\omega$ terms on both sides.

$$A - B = 10$$

$$\frac{-2B}{3} - B = 10$$

$$\frac{-5B}{3} = 10$$

$$B = -6$$

Consider,

$$\begin{aligned} A &= \frac{-2B}{3} \\ &= \frac{-2(-6)}{3} \\ &= \frac{12}{3} \\ &= 4 \end{aligned}$$

Step 7 of 12

Consider,

$$\begin{aligned} \left[\frac{10j\omega}{(2-j\omega)(j\omega+3)} \right] &= \frac{A}{(2-j\omega)} + \frac{B}{(j\omega+3)} \\ &= \frac{4}{(2-j\omega)} - \frac{6}{(j\omega+3)} \end{aligned}$$

Apply Inverse Fourier transform.

$$\begin{aligned} F^{-1}[F(\omega)] &= F^{-1} \left[\frac{4}{(2-j\omega)} - \frac{6}{(j\omega+3)} \right] \\ &= F^{-1} \left[\frac{4}{(2-j\omega)} \right] - F^{-1} \left[\frac{6}{(j\omega+3)} \right] \end{aligned}$$

Step 8 of 12

Consider,

$$\begin{aligned} F^{-1}\left[\frac{4}{(2-j\omega)}\right] &= 4\left(F^{-1}\left[\frac{1}{(2-j\omega)}\right]\right) \\ &= 4e^{2t}u(-t) \end{aligned}$$

And,

$$\begin{aligned} F^{-1}\left[\frac{6}{(j\omega+3)}\right] &= 6\left(F^{-1}\left[\frac{1}{(j\omega+3)}\right]\right) \\ &= 6e^{-3t}u(t) \end{aligned}$$

$$\begin{aligned} f(t) &= F^{-1}\left[\frac{4}{(2-j\omega)}\right] - F^{-1}\left[\frac{6}{(j\omega+3)}\right] \\ &= 4e^{2t}u(-t) - 6e^{-3t}u(t) \end{aligned}$$

Therefore, the Inverse Fourier transform of $F(\omega)$ is $\boxed{4e^{2t}u(-t) - 6e^{-3t}u(t)}$.

Step 9 of 12

(c)

Consider the Fourier transform $H(\omega)$ as,

$$H(\omega) = \frac{60}{(-\omega^2 + j40\omega + 1300)}$$

Determine the inverse Fourier transform of $H(\omega)$.

$$\begin{aligned} F^{-1}[F(\omega)] &= F^{-1}\left[\frac{60}{(-\omega^2 + j40\omega + 1300)}\right] \\ \left[\frac{60}{(-\omega^2 + j40\omega + 1300)}\right] &= \frac{60}{(j\omega)^2 + j40\omega + 900} \\ &= \frac{60}{(j\omega + 20)^2 + 900} \end{aligned}$$

Step 10 of 12

Consider,

$$\begin{aligned} F^{-1}\left[\frac{b}{(j\omega+a)^2+b^2}\right] &= 2e^{-at}\sin(bt)u(t) \\ F^{-1}\left[\frac{60}{(-\omega^2 + j40\omega + 1300)}\right] &= 2\left[F^{-1}\left[\frac{30}{(j\omega+20)^2+900}\right]\right] \\ h(t) &= 2e^{-20t}\sin(30t)u(t) \end{aligned}$$

Therefore, the Inverse Fourier transform $H(\omega)$ is $\boxed{2e^{-20t}\sin(30t)u(t)}$.

Step 11 of 12

(d)

Consider the Fourier transform $Y(\omega)$ as,

$$Y(\omega) = \frac{\delta(\omega)}{(j\omega+1)(j\omega+2)}$$

Determine the inverse Fourier transform of $Y(\omega)$.

$$\begin{aligned} F^{-1}[Y(\omega)] &= F^{-1}\left[\frac{\delta(\omega)}{(j\omega+1)(j\omega+2)}\right] \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \left[\frac{\delta(\omega)}{(j\omega+1)} - \frac{\delta(\omega)}{(j\omega+2)}\right] e^{j\omega t} dt \end{aligned}$$

Step 12 of 12

Apply sifting property.

$$\int_{-\infty}^{+\infty} [\delta(\omega - \omega_o) X(\omega)] = X(\omega_o)$$

$$y(t) = \frac{1}{2\pi} \left[\frac{1}{(j(0)+1)} - \frac{1}{(j(0)+2)} \right]$$

$$\begin{aligned} y(t) &= \frac{1}{2\pi} \left(1 - \frac{1}{2} \right) \\ &= \frac{1}{2\pi} \left(\frac{1}{2} \right) \\ &= \frac{1}{4\pi} \end{aligned}$$

Therefore, the Fourier transform of $Y(\omega)$ is $\boxed{\frac{1}{4\pi}}$.